Fiscal Year:	FY 2014	**Task Last Updated:**	FY 12/01/2014
PI Name:	Paul, Anna-Lisa Ph.D.		
Project Title:	Cell Signaling in Undifferentiated Cells - Perceiving the Environment without Specialized Tissues		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Cell Culture (2) Bioregenerative Life Support		
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Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	09/01/2012	**End Date:**	08/31/2014
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ferl, Robert (University of Florida)		
Grant/Contract No.:	NNX12AK80G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of our program is to understand fundamental molecular biological responses to extraterrestrial environments. This goal aligns with Space Biology goals to understand the impact of spaceflight on biological systems to better enable the exploration imperative. Within that context, the goal of this proposal is to explore the mechanism by which undifferentiated cells, cells presumably lacking the specializations typical of gravity response mechanisms, perceive the novel environment of spaceflight. We used undifferentiated callus cells from Arabidopsis to reveal unique gene expression patterns in response to true spaceflight as part of the BRIC-16 experiment in 2010. We followed up with examination of the response of these cells to a variety of terrestrial environments and stimuli, including spaceflight analogs such as clinorotation, hyper-g centrifugation, and parabolic flight. The present proposal moves beyond phenomenon observation to address specific hypotheses derived from these initial spaceflight and ground experiments. In BRIC-16 biologically replicated DNA microarray and RNA digital transcript profiling revealed several hundred genes in seedlings and cell cultures that are significantly affected by launch and spaceflight. While the response in intact Arabidopsis seedlings was significant but moderate in intensity, the response of undifferentiated cell cultures was dramatic, intense, and in completely different gene sets than in the intact seedlings. That undifferentiated plants cells uniquely perceive gravity raises powerful questions of gravity perception and response, and sets the stage for hypothesis-driven experimentation directed toward understanding this perception and response. We hypothesize that the response of cell cultures is not driven by the same gravity response mechanisms known in intact plants. We will use culture lines derived from gravity signaling mutants to test this hypothesis. This will investigate the underlying mechanisms by which undifferentiated cells detect and respond to the spaceflight environment in the absence of specialized tissue or organized developmental structures known to detect gravity. The read-out data will again be in the form of gene expression patterns that can be directly compared to our previous flight and ground data. These experiments are carefully scaled to the BRIC operational context and enable us to dig deeply and specifically into mechanisms associated with plant responses to the novel environment of spaceflight. Gaining insights into such mechanisms is recognized as fundamental within the previous decadal study and underpin answers to some of the biggest questions in spaceflight plant biology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding fundamental processes in gene expression and signal transduction associated with adaptation to novel environments.
Task Progress:	The BRIC-17 experiment led to some remarkable scientific developments. First, it showed that undifferentiated cells do respond to microgravity, can sense and recognize the sub-g environment and adjust the cellular processes to continue to perform in the new gravitational condition. The BRIC-17 experiment gave an insight as of how mutated cells, very different on the ground, also coped with the microgravity environment. Second, the BRIC-17 experiment demonstrated that standard approaches of comparing the genes expression in the treated sample relative to the untreated should be taken with great caution, particularly if the mutant cell lines are part of the study. The WT and mutant comparisons with a standard analysis approach looked fundamentally different, yet there were hidden similarities. The novel and original approach of expanding the comparison across the ground or the spaceflight gene expression level with multiple cross-comparisons resulted in a very different conclusion. The cells shared many genes for which expression was brought to the same level, and which could possibly reflect the required changes to cellular processes in microgravity. However, the way each cell line achieved the even expression of specific genes was very different. Therefore the two very distinctive terms were coined, the response to microgravity and the state in microgravity. The response to microgravity was considered as a pool of all genes that expression changed in orbit as compared to the respective ground controls. The microgravity state was considered as a specific level of some genes expression, likely required to readjust structures and processes to the new gravitational filed, so the new order and homeostasis required in the microgravity environment could be achieved. Assuming that being in microgravity would require some structural and physiological changes in the cells, they would be attained but in different ways depending on cell types. There are three manuscripts in preparation derived from this research, which will likely be published in mid to late 2015.
Bibliography Type:	Description: (Last Updated: 11/20/2025)
Articles in Peer-reviewed Journals	Ferl RJ. "Suborbital vehicles to study transition adaptation to spaceflight – Why biologists should care about the new suborbital flight opportunities." Gravit Space Res. 2014 Dec;2(2):58-65. http://gravitationalandspacebiology.org/index.php/journal/article/view/658 , Dec-2014
Articles in Peer-reviewed Journals	Zupanska AK, Schultz ER, Yao J, Sng NJ, Zhou M, Callaham JB, Ferl RJ, Paul A-L. "ARG1 functions in the physiological adaptation of undifferentiated plant cells to spaceflight." Astrobiology. 2017 Nov;17(11):1077-111. Epub 2017 Oct 31. https://doi.org/10.1089/ast.2016.1538 ; PubMed PMID: 29088549 , Nov-2017
Articles in Peer-reviewed Journals	Zupanska AK, LeFrois C, Ferl RJ, Paul A-L. "HSFA2 functions in the physiological adaptation of undifferentiated plant cells to spaceflight." Int J Mol Sci. 2019 Jan 17;20(2):E390. https://doi.org/10.3390/ijms20020390 ; PubMed PMID: 30658467; PubMed Central PMCID: PMC6359015 , Jan-2019